

In-term Test Q4 2025-2026

Student's Name	Student's ID	Github Repository Link
Thân Quế Ngọc	2232300060	View on Github

*This paper consists of **SIX** practice questions, carrying a total of 100 marks. Questions 1 and 2 are worth 10 marks each, while questions 3 to 6 carry 20 marks each.*

Students have 90 minutes to answer all the questions. No extra time is given for reading the questions. Additional Exam Booklet is provided on Moodle. Students use computers with MySQL or SQL Server to complete the questions.

Students must submit the Exam Answer via Moodle. (script .sql and report as .pdf)

File Name format: MIS443_ID_FullName.zip.

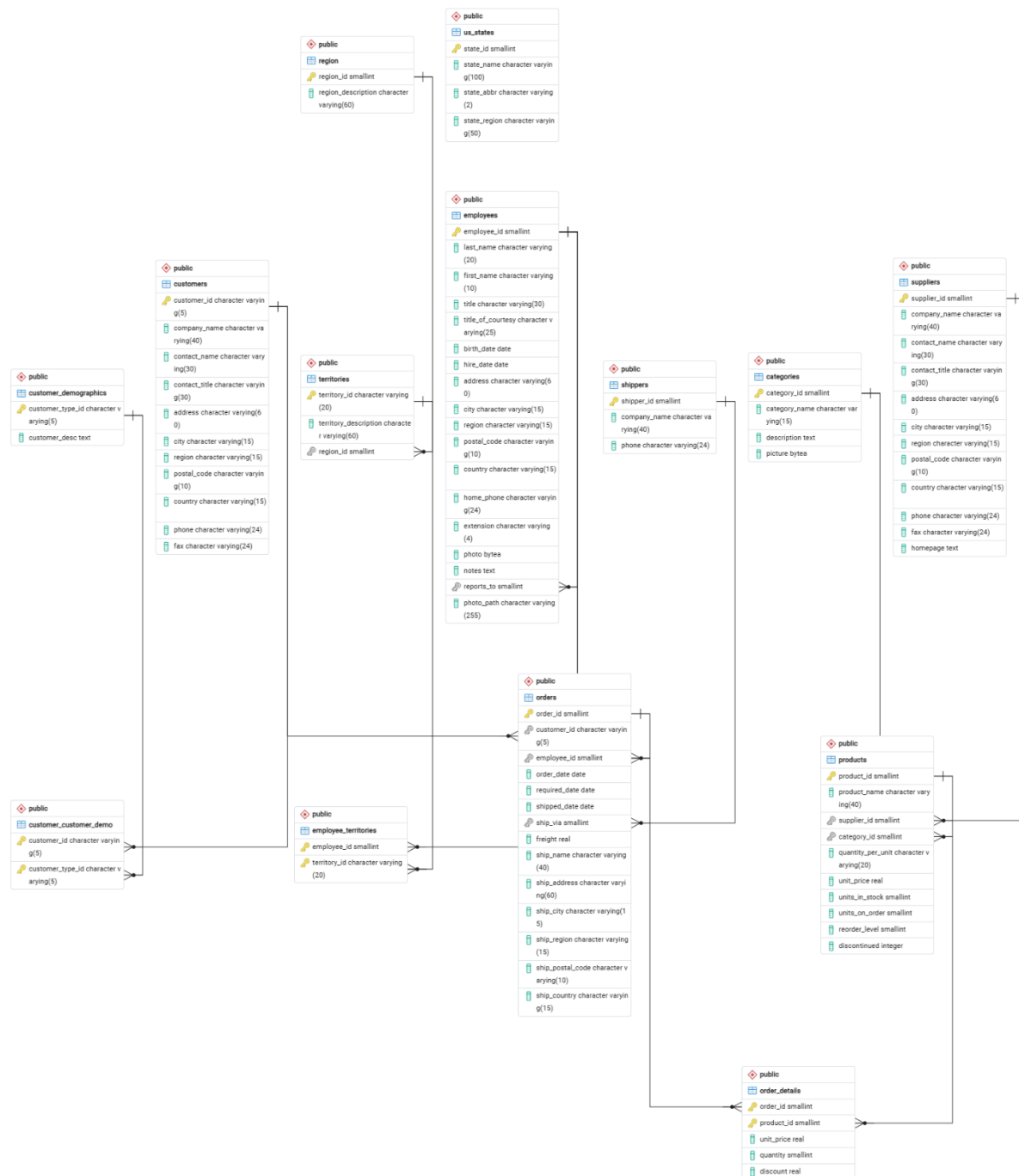
Scenario:

This dataset represents Northwind Traders, a mid-sized international trading company that sells food and beverage products to business customers. The database captures essential information about customers, orders, products, and shipping activities. The primary goal of this database is to support data-driven insights into customer purchasing behavior, order activity, and delivery operations, enabling the company to improve sales performance, optimize logistics, and make informed operational decisions.

- ✓ Customers: Stores each customer's unique ID and company information (e.g., company name, contact details, location). This table supports customer segmentation and relationship management.
- ✓ Orders: Records each order placed by a customer, including order date, shipping date, and shipping method. It provides the foundation for analyzing order frequency and delivery performance.
- ✓ Order_Details: Contains line-item details for each order (products, quantities, unit prices, and discounts). This table supports transaction-level analysis of product demand and sales patterns.
- ✓ Products: Lists all products sold by Northwind, including product name and related product attributes. It is used to analyze demand across different products.
- ✓ Shippers: Contains shipping partner information used to deliver orders. This table supports evaluation of logistics and shipping performance.

In addition, students must create a new table named “students”. This table should include three attributes: studentid, fullname, and email. You may decide the appropriate data types and length for each attribute.

Note: The SQL script for loading the Northwind tables (include 14 tables) is available on Moodle. Only the “students” table must be created entirely by the student.



ER Model (five main tables)

Question 1 (10 marks): Database Setup (10 marks)

Using PostgreSQL and pgAdmin:

- Create a database named yourfullname. Then move all Northwind tables into this schema. (5 marks)

(b) Create a new table called students inside schema exam with the following columns:

Studentid: 5-digit number, Primary Key

Fullname: Required

Email: Must be unique

Insert your own information into the table. (5 marks).

-- Test results

SELECT table_name, column_name, data_type

FROM information_schema.columns

WHERE table_schema = 'public'

ORDER BY table_name, ordinal_position;

Sample Output:

Results 1: Verify result from load and create database

The screenshot shows the pgAdmin 4 interface. The query editor on the right contains the following SQL query:

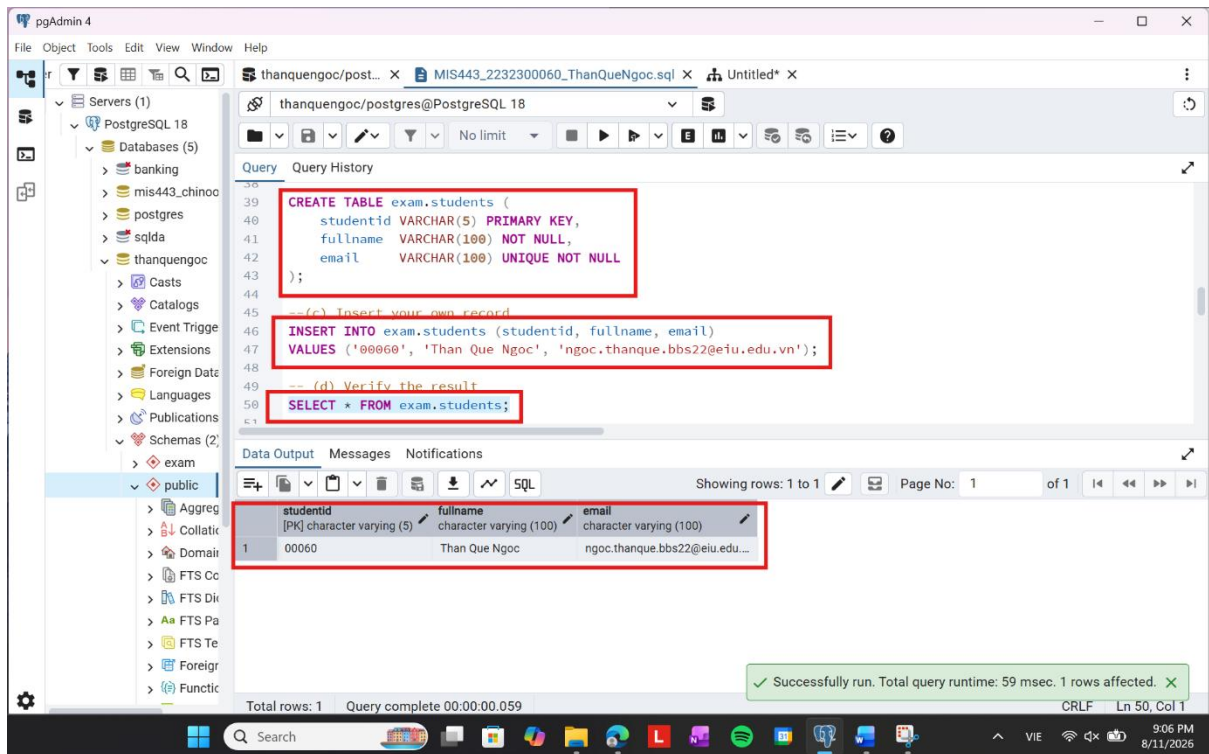
```
SELECT table_name, column_name, data_type
FROM information_schema.columns
WHERE table_schema = 'public'
ORDER BY table_name, ordinal_position;
```

The query has been executed successfully, and the results are displayed in the Data Output pane at the bottom. The results show a list of tables and their columns in the 'public' schema.

table_name	column_name	data_type
categories	category_id	smallint
categories	category_name	character varying
categories	description	text
categories	picture	bytea
customer_customer_de...	customer_id	character varying
customer_customer_de...	customer_type_id	character varying
customer demographics	customer tnoe id	character varvino

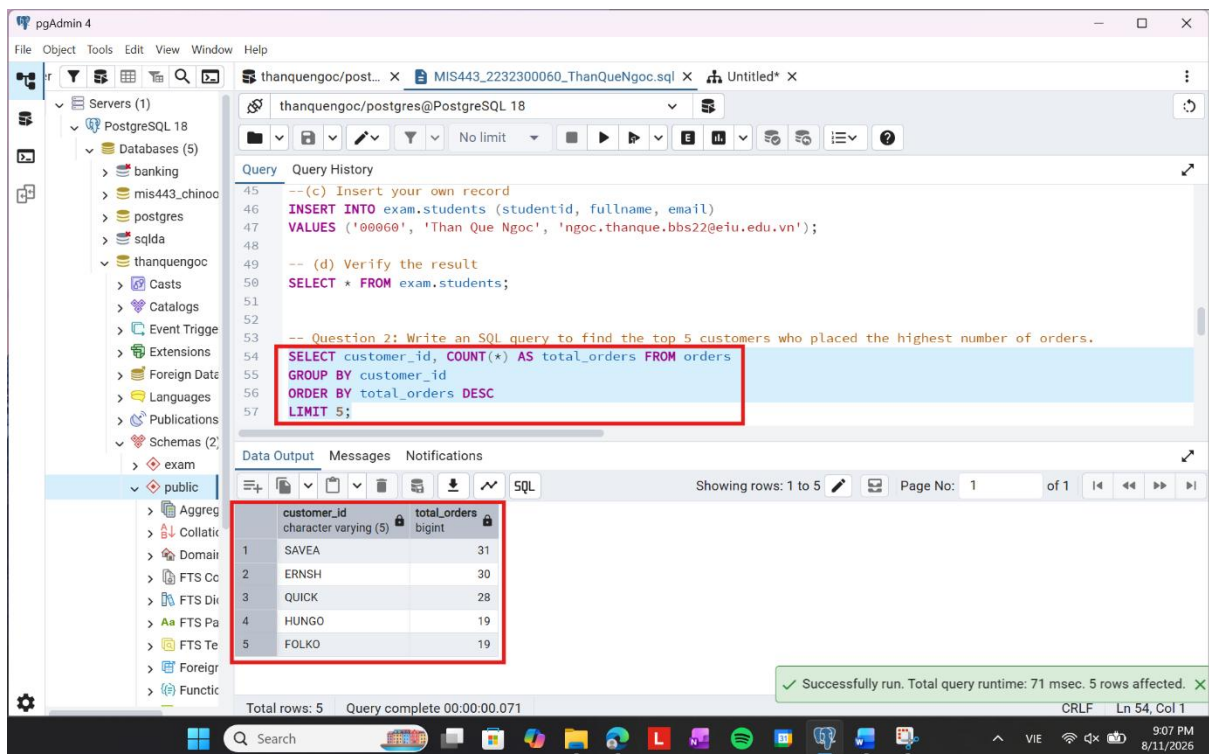
The status bar at the bottom indicates: "Total rows: 92 Query complete 00:00:00.063". A green message box at the bottom right says: "Successfully run. Total query runtime: 63 msec. 92 rows affected."

Results 2: Verify result from table student



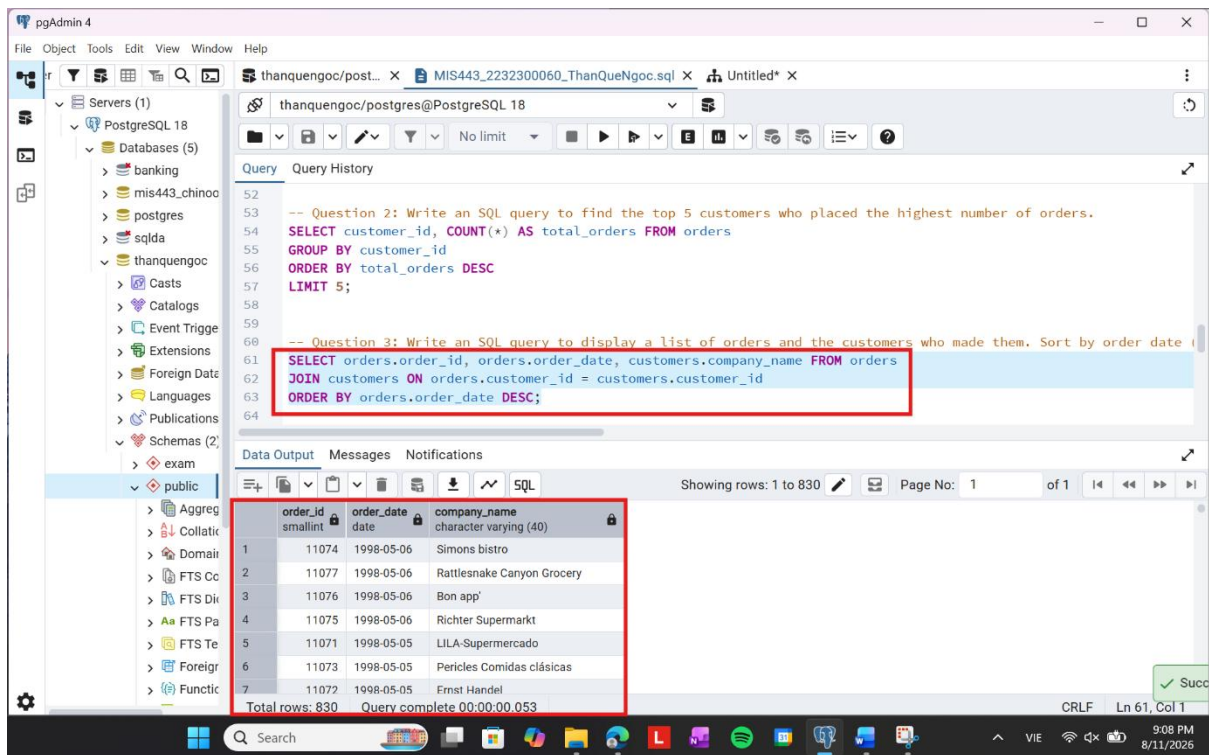
Question 2 (10 marks): Write an SQL query to find the top 5 customers who placed the highest number of orders.

Expected Output:



Question 3 (20 marks): Write an SQL query to display a list of orders and the customers who made them. Sort by order date (newest first).

Expected Output:



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including the 'public' schema. The main query editor contains two SQL queries. The first query is highlighted with a red box and is as follows:

```
-- Question 3: Write an SQL query to display a list of orders and the customers who made them. Sort by order date (
SELECT orders.order_id, orders.order_date, customers.company_name FROM orders
JOIN customers ON orders.customer_id = customers.customer_id
ORDER BY orders.order_date DESC;
```

The second query is also highlighted with a red box and is as follows:

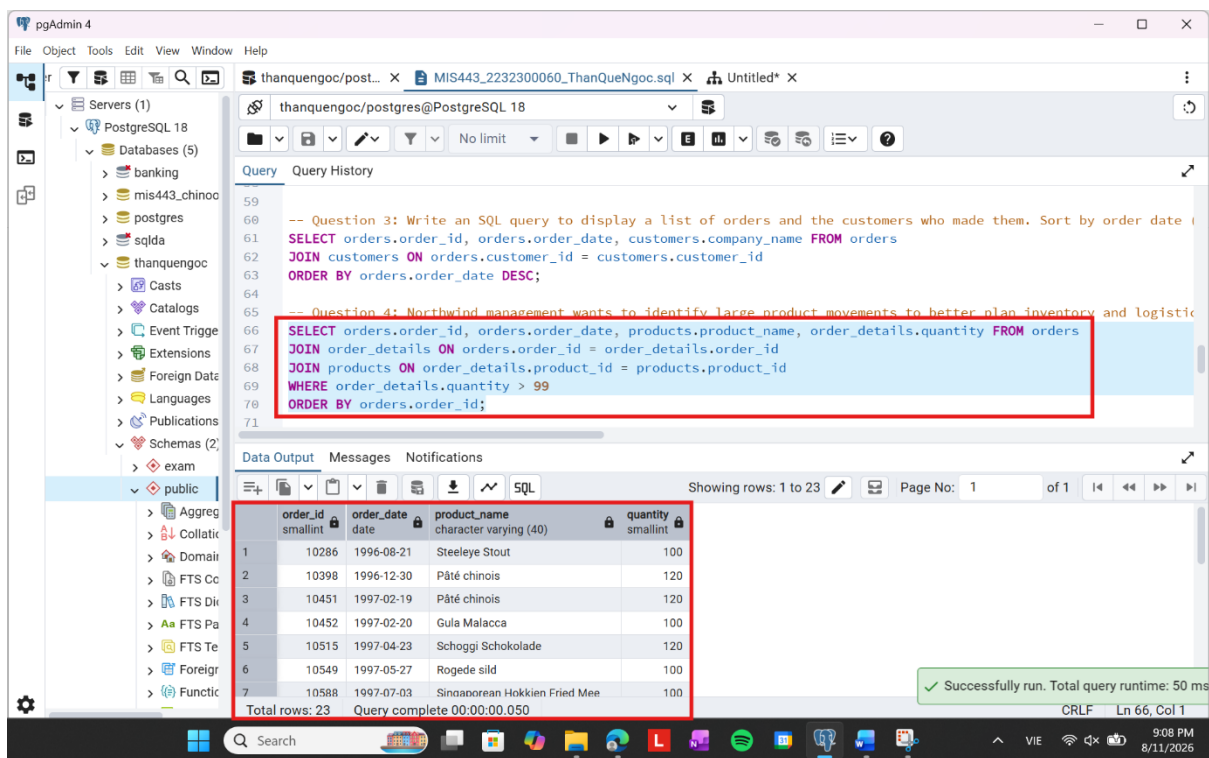
```
-- Question 4: Northwind management wants to identify large product movements to better plan inventory and logistic
SELECT orders.order_id, orders.order_date, products.product_name, order_details.quantity FROM orders
JOIN order_details ON orders.order_id = order_details.order_id
JOIN products ON order_details.product_id = products.product_id
WHERE order_details.quantity > 99
ORDER BY orders.order_id;
```

The 'Data Output' pane shows the results of the first query, which are displayed in a table with the following columns: order_id, order_date, and company_name. The table contains 7 rows of data, with the first row being 11074, 1998-05-06, and Simons bistro. The total number of rows is 830.

order_id	order_date	company_name	
1	11074	1998-05-06	Simons bistro
2	11077	1998-05-06	Rattlesnake Canyon Grocery
3	11076	1998-05-06	Bon app'
4	11075	1998-05-06	Richter Supermarkt
5	11071	1998-05-05	LILA-Supermercado
6	11073	1998-05-05	Pericles Comidas clásicas
7	11072	1998-05-05	Ernst Handel

Question 4 (20 marks): Northwind management wants to identify large product movements to better plan inventory and logistics. Write an SQL query to display orders where a product was purchased in large quantity (more than 99 units in a single order).

Expected Output:



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including the 'public' schema. The main query editor contains two SQL queries. The first query is highlighted with a red box and is as follows:

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-- Question 3: Write an SQL query to display a list of orders and the customers who made them. Sort by order date (
SELECT orders.order_id, orders.order_date, customers.company_name FROM orders
JOIN customers ON orders.customer_id = customers.customer_id
ORDER BY orders.order_date DESC;
```

The second query is also highlighted with a red box and is as follows:

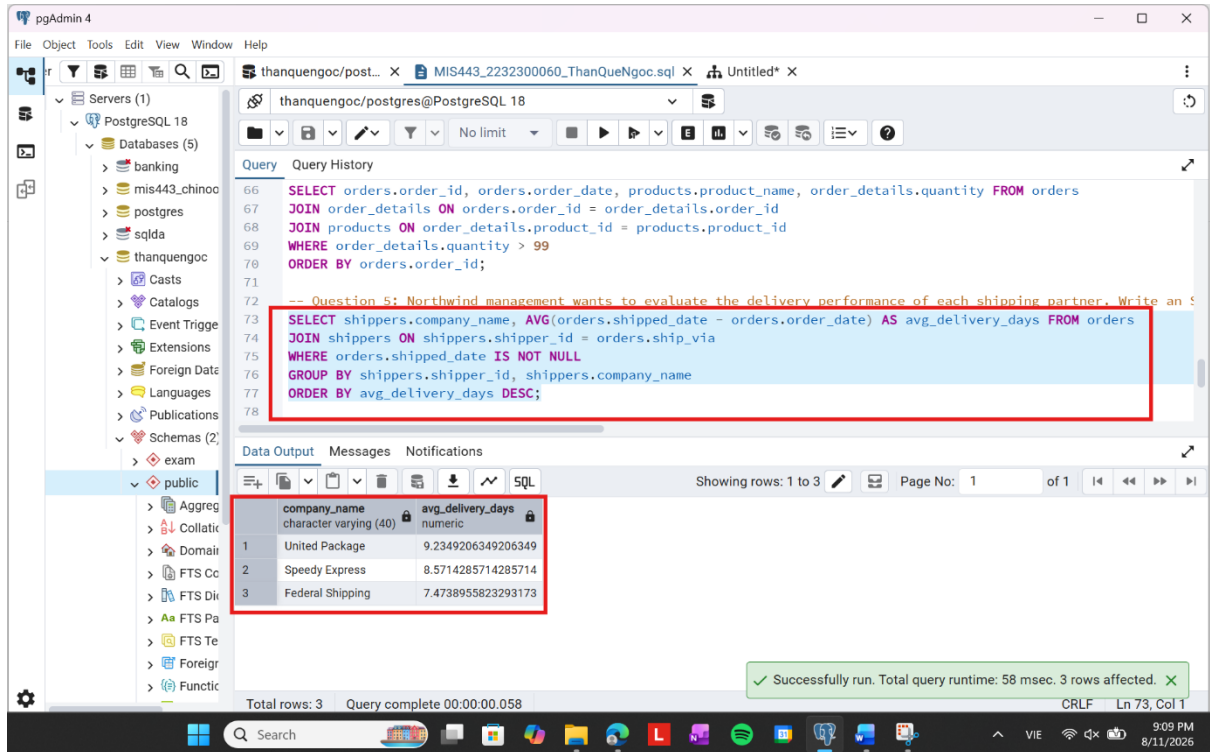
```
-- Question 4: Northwind management wants to identify large product movements to better plan inventory and logistic
SELECT orders.order_id, orders.order_date, products.product_name, order_details.quantity FROM orders
JOIN order_details ON orders.order_id = order_details.order_id
JOIN products ON order_details.product_id = products.product_id
WHERE order_details.quantity > 99
ORDER BY orders.order_id;
```

The 'Data Output' pane shows the results of the first query, which are displayed in a table with the following columns: order_id, order_date, and company_name. The table contains 7 rows of data, with the first row being 11074, 1998-05-06, and Simons bistro. The total number of rows is 830.

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6	11073	1998-05-05	Pericles Comidas clásicas
7	11072	1998-05-05	Ernst Handel

Question 5 (20 marks): Northwind management wants to evaluate the delivery performance of each shipping partner. Write an SQL query to calculate the average delivery time (in days) for each shipper. Delivery time = shipped_date – order_date

Expected Output:



The screenshot shows the pgAdmin 4 interface. The query editor contains the following SQL query:

```

SELECT orders.order_id, orders.order_date, products.product_name, order_details.quantity FROM orders
JOIN order_details ON orders.order_id = order_details.order_id
JOIN products ON order_details.product_id = products.product_id
WHERE order_details.quantity > 99
ORDER BY orders.order_id;

-- Question 5: Northwind management wants to evaluate the delivery performance of each shipping partner. Write an SQL query to calculate the average delivery time (in days) for each shipper. Delivery time = shipped_date – order_date
SELECT shippers.company_name, AVG(orders.shipped_date - orders.order_date) AS avg_delivery_days FROM orders
JOIN shippers ON shippers.shipper_id = orders.ship_via
WHERE orders.shipped_date IS NOT NULL
GROUP BY shippers.shipper_id, shippers.company_name
ORDER BY avg_delivery_days DESC;

```

The query results are displayed in the Data Output tab, showing 3 rows:

	company_name	avg_delivery_days
1	United Package	9.2349206349206349
2	Speedy Express	8.5714285714285714
3	Federal Shipping	7.4738955823293173

At the bottom, a status bar indicates: "Successfully run. Total query runtime: 58 msec. 3 rows affected."

Question 6 (20 marks): Northwind wants to identify the most active customers (customers who place orders most frequently) to target retention campaigns. Write an SQL query to rank customers based on their total number of orders (highest = rank 1). Customers with the same number of orders must have the same rank.

Expected Output:

pgAdmin 4

File Object Tools Edit View Window Help

thanquengoc/post... X MIS443_2232300060_ThanQueNgoc.sql X Untitled* X

thanquengoc/postgres@PostgreSQL 18

Query Query History

```

77 ORDER BY avg_delivery_days DESC;
78
79
80 -- Question 6: Northwind wants to identify the most active customers (customers who place orders most frequently)
81
82 SELECT customers.customer_id,
83        customers.company_name,
84        COUNT(orders.order_id) AS total_orders,
85        RANK() OVER (ORDER BY COUNT(orders.order_id) DESC) AS customer_rank
86 FROM customers
87 JOIN orders ON customers.customer_id = orders.customer_id
88 GROUP BY customers.customer_id, customers.company_name
89 ORDER BY customer_rank;

```

Data Output Messages Notifications

Showing rows: 1 to 89 Page No: 1 of 1

	customer_id [PK] character varying (5)	company_name character varying (40)	total_orders bigint	customer_rank bigint
1	SAVEA	Save-a-lot Markets	31	1
2	ERNSH	Ernst Handel	30	2
3	QUICK	QUICK-Stop	28	3
4	HUNGO	Hungry Owl All-Night Grocers	19	4
5	FOLKO	Folk och få HB	19	4
6	BERGS	Berglunds snabbköp	18	6
7	HII AA	HII ARION-Abastos	18	6

Total rows: 89 Query complete 00:00:00.058

CRLF Ln 81, Col 1

9:10 PM 8/11/2026

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